COURSE NAME

DISCRETE MATHEMATICS

CSC 1204

(UNDERGRADUATE)

CHAPTER 9

MATRICES

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MATRIX

- A matrix is a rectangular array of numbers. A matrix with m rows and n columns is called an $m \times n$ matrix.
- The plural of **matrix** is **matrices**
- A matrix with the same number of rows as columns is called a square (or square matrix)
- Two matrices are equal if they have the same number of rows and the same number of columns and the corresponding entries in every position are equal.

$$\begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 3 \end{bmatrix}$$ is a 3×2 matrix

MATRIX ARITHMETIC: ADDITION OF TWO MATRICES

- Let A= $[a_{ij}]$ and B=$[b_{ij}]$ be $m \times n$ matrices. The sum of A and B, denoted by A+B, is the $m \times n$ matrix that has $a_{ij}+b_{ij}$ as its (i,j)th element. In other words , $A + B = [a_{ij} + b_{ij}]$
- The sum of two matrices of the same size is obtained by adding elements in the corresponding positions.
- The subtraction of two matrices of the same size is obtained by subtracting elements in the corresponding positions
- Only matrices with the same dimensions can be added and subtracted. The resulting matrix has the same dimension of the two matrices being added or subtracted.
- *Note*: Matrices of different sizes cannot be added/subtracted

1. $$\begin{bmatrix} -2 & 0 & 4 \\ 3 & -10 & 12 \\ 3 & -2 & -2 \end{bmatrix} + \begin{bmatrix} -4 & 6 & 0 \\ -15 & 2 & -4 \\ 6 & 7 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -2-4 & 0+6 & 4+0 \\ 3-15 & -10+2 & 12-4 \\ 3+6 & -2+7 & -2+1 \end{bmatrix} \Rightarrow \begin{bmatrix} -6 & 6 & 4 \\ -12 & -8 & 8 \\ 9 & 5 & -1 \end{bmatrix}$$

MATRIX ARITHMETIC: ADDITION OF TWO MATRICES

Q: Find out the sum of the following two matrices A and B.

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & -3 \\ 3 & 4 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 4 & -1 \\ 1 & -3 & 0 \\ -1 & 1 & 2 \end{bmatrix} \qquad A + B = \begin{bmatrix} 4 & 4 & -2 \\ 3 & -1 & -3 \\ 2 & 5 & 2 \end{bmatrix}$$

MATRIX ARITHMETIC: PRODUCT OF TWO MATRICES

- Let **A** be $m \times k$ matrix and **B** be $k \times n$ matrix. The product of **A** and **B**, denoted by **AB**, is the $m \times n$ matrix with its (i,j)th element equal to the sum of the products of the corresponding entries from the ith row of **A** and the jth column of **B**.
- In other words, If **AB** = $[c_{ij}]$, then $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{ik}b_{kj}$.
- *Note*: A product of two matrices is defined only when the number of columns in the first matrix *equals* the number of rows of the second matrix.

Q: Find the product AB from the following matrices, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -2 \\ 5 & 3 & -1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 6 & 3 \\ 4 & -3 \\ -7 & 2 \end{bmatrix} = \begin{bmatrix} 20 & -1 \\ 49 & 4 \end{bmatrix}$$

- 1(6) + 0(4) + (-2)(-7) = 6 + 0 +14 = 20
- 1(3) + 0(-3) + (-2)(2) = 3 – 0 – 4 = -1
- 5(6)+3(4)+(-1)(-7) = 30 + 12 + 7 = 49
- 5(3)+3(-3)+(-1)(2) = 15 – 9 – 2 = 4

ZERO-ONE MATRICES

- A matrix with entries that are either 0 or 1 is called *zero-one matrix*.
- Algorithms operating on discrete structures represented by zero-one matrices are based on Boolean arithmetic defined by the following Boolean operations:

$$b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = b_2 = 1 \\ 0 & \text{otherwise} \end{cases} \qquad b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

- Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be an $m \times n$ zero-one matrices.
- The ***join*** of **A** and **B** is the zero-one matrix with (i,j)th entry $a_{ij} \vee b_{ij}$. The *join* of **A** and **B** is denoted by $\mathbf{A} \vee \mathbf{B}$.
- The **meet** of **A** and **B** is the zero-one matrix with (i,j)th entry $a_{ij} \wedge b_{ij}$. The *meet* of **A** and **B** is denoted by $\mathbf{A} \wedge \mathbf{B}$.
- **Note:** These operations(join and meet) are only possible when **A** and **B** have the same size, just as in the case of matrix addition.

JOIN AND MEET OF ZERO-ONE MATRICES

Q: Find the **join** and **meet** of the zero-one matrices **A** and **B**.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution:

The join of **A** and **B** is

$$\mathbf{A} \vee \mathbf{B} = \begin{bmatrix} 1 \vee 0 & 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 1 \vee 1 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

The meet of **A** and **B** is

$$\mathbf{A} \wedge \mathbf{B} = \begin{bmatrix} 1 \wedge 0 & 0 \wedge 1 & 1 \wedge 0 \\ 0 \wedge 1 & 1 \wedge 1 & 0 \wedge 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

BOOLEAN PRODUCT OF ZERO-ONE MATRICES

- Let $\mathbf{A} = [a_{ij}]$ be an $m \times k$ zero-one matrix and $\mathbf{B} = [b_{ij}]$ be a $k \times n$ zero-one matrix.
- The *Boolean product* of **A** and **B**, denoted by $\mathbf{A} \odot \mathbf{B}$, is the $m \times n$ zero-one matrix with (i,j)th entry c_{ij} where $c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \ldots \vee (a_{ik} \wedge b_{kj})$

Q: Find the Boolean product of **A** and **B**, where $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

$$\mathbf{A} \odot \mathbf{B} = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix}$$

$$= \begin{bmatrix} 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \\ 0 \vee 0 & 0 \vee 1 & 0 \vee 1 \\ 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

REFERENCES

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- *Discrete and Combinatorial Mathematics(3rd Edition)* by *R. P. Grimaldi,*, published by Addison-Wesley